

Domingo Ranieri

AI Engineer | LLM Applications, RAG, Agents & MCP

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Profile

AI-focused software engineer with hands-on experience building LLM-powered applications, RAG pipelines, AI agents, and MCP integrations. I design, test, and deploy end-to-end AI systems using Python, FastAPI, Docker, and Kubernetes, with a strong emphasis on code quality, automated testing, and production readiness.

My core work spans retrieval architecture, prompt engineering, evaluation frameworks, and secure LLM platform design. I use AI-assisted development tools — including Claude Code and Codex — as part of my daily workflow, and I actively prototype new ideas driven by a genuine passion for building and experimenting with emerging AI technologies.

I hold a Master's degree in Theoretical Physics (110/110 cum laude), which gave me a rigorous analytical foundation and the mathematical grounding behind machine learning algorithms.

Education

Master's Degree in Physics

Theoretical Physics

Final grade: 110/110 cum laude

Thesis: Simulation of neuromuscular control using a quantum computer

University of Bologna

2020 - 2022

Bachelor's Degree in Physics

Physics

Final grade: 110/110 cum laude

Thesis: Fractal Universe model and Cosmic Acceleration

University of Bologna

2017 - 2020

Work experience

ICSC

Software Engineer

Mar 2025 - Present

Bologna, Italy

- Design and develop AI-powered applications and automation tools
- Build and maintain containerised services with **Python**, **FastAPI**, **Docker**, and **Kubernetes**, following CI/CD practices with **GitHub Actions** and automated testing with **pytest**
- Deliver technical courses on Machine Learning, and data engineering to internal and external audiences
- Contribute to collaborative research projects and present findings at national conferences

ICSC-INFN

Research Fellow

Sep 2023 - Feb 2025

Bologna, Italy

- Designed and developed AI-based services and distributed system prototypes using **Python**, **Docker**, and cloud-native architectures
- Implemented blockchain-based solutions for data integrity and traceability in research environments
- Participated in international workshops and contributed to cross-team technical documentation

Key Projects

- Built an end-to-end **RAG system** for question answering over custom document collections, using **Milvus** as the vector database and a fully **local inference pipeline** (Ollama + Hugging Face) with no external API dependencies
- Designed a **context-aware chunking strategy** to preserve semantic coherence and document-level context during embedding and retrieval
- Implemented and benchmarked **dense and hybrid retrieval pipelines**, achieving best performance with hybrid retrieval (MRR = 0.90, Recall@1 = 0.74, Recall@5 = 0.89)
- Developed an **LLM-as-a-judge evaluation framework** to assess answer quality beyond retrieval metrics, scoring correctness, relevance, and completeness
- Architected the system to be **modular and API-agnostic**, allowing seamless integration with providers such as OpenAI or Anthropic
- Ensured code reliability through **unit testing with pytest** and automated **CI/CD pipelines via GitHub Actions**

- Developed a **FastAPI-based chatbot hosting platform** enabling secure interaction with both public and private **MCP (Model Context Protocol) servers**
- Implemented security controls following the **OWASP MCP Security Cheat Sheet**: controlled tool exposure, role-based access restriction, and secure request handling
- Integrated **Keycloak (OAuth2)** for authentication and authorization, **PostgreSQL** for persistent storage, and **Redis** for caching and session optimization
- Deployed using a fully **Dockerized architecture** designed to be Kubernetes-ready for scalable deployment, with a locally hosted **Ollama LLM runtime** for private inference
- Validated functionality through **unit testing with pytest** and continuous integration via **GitHub Actions**

Technical skills

Programming	Python, Go, Bash, C++
ML/AI	Scikit-learn, TensorFlow, PyTorch, Keras, Ollama, vLLM, LangGraph, LangChain, CrewAI, AWS Bedrock
Data Engineering	Kafka, Airflow, MinIO
Cloud & DevOps	Docker, Kubernetes, Ansible
Databases	PostgreSQL, MongoDB, MySQL, Redis, Milvus
Web	Flask, FastAPI, HTML, CSS, JavaScript, Streamlit
Tools	Claude Code, Codex, MLflow, Grafana, Prometheus, Loki, Git, Keycloak

Languages

Italian	Native
English	C1

Communication and Interpersonal Skills

- Effective presentation and public speaking
 - Strong technical writing and documentation
- Collaborative teamwork in multidisciplinary environments
 - Ability to explain complex concepts clearly